

PocketSmart AI

Date	15 March 2026
Team ID	xxxxxx
Project Name	PocketSmart AI
Maximum Marks	4 Marks

Solution Requirements

The following requirements are prepared for the PocketSmart AI concept. The template distinguishes Functional Requirements (what the system does) from Non-Functional Requirements (how well it operates).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely access PocketSmart and provide the information needed for a planning session. For a prototype, authentication can be lightweight; production use should use secure identity verification.	Medium
2	Authorization levels	The system should distinguish normal users from administrative/project-management access if administration is introduced. Normal users manage only their own planning inputs and results.	Medium
3	External interfaces	Connect to external product/service sources or APIs to retrieve relevant product, service, price and availability information for recommendations.	High
4	Transactions processing	Process the user's budget, category, preferences and constraints; calculate budget allocation and rank suitable recommendations. No payment transaction is required for the core prototype.	High
5	Reporting	Provide a planning summary showing budget, recommended allocation, shortlisted options and estimated spend. Allow the user to review the recommendation result.	Medium
6	Business rules, etc.	Recommendations must respect the user's stated budget and preferences. Results should be personalized by category, style/occasion and other supplied constraints.	High
7	Compliance to laws or regulations	Protect user-provided personal and preference data. Apply applicable privacy and data-protection requirements for the deployment location. Specific regulatory certification is not defined by the project source.	Medium
8	Other	Support three initial use cases: Home Interior Planning, Party Budget Planning and Jewelry Recommendations. The design should allow additional categories later.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Recommendation requests should return results promptly enough for interactive planning.	Prototype target: typical recommendation response within 5 seconds under normal load.
2	Scalability	The architecture should support additional users, recommendation categories and external sources without redesigning the complete solution.	Modular services and APIs; capacity to be increased as usage grows.
3	Security & Data Privacy	Protect user inputs, preferences and any stored data; use secure transport and avoid exposing sensitive information in logs.	HTTPS/TLS in deployment; least-privilege access; no sensitive data in application logs.
4	Reliability & Availability	The system should handle temporary external-source or AI-service failures gracefully.	Show a useful error/fallback message and preserve the user's request; target 99% prototype service availability.
5	Usability & Accessibility	The interface should be simple for users to enter a budget and preferences and understand recommendations.	Responsive interface; clear labels; keyboard-friendly controls; readable recommendation summaries.
6	Other	Maintainability and extensibility should allow new planning categories and recommendation providers to be added.	Modular category/rule design and documented APIs/components.